Lesson 7

Tables and Data Integrity Enforcement

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Tables

- Table types:
  - Permanent (CREATE TABLE t)
  - Temporary (USE Tempdb CREATE TABLE t)
    - Accessible by all user at any connection
    - Deleted when server restarts
  - Local temporary (CREATE TABLE #t)
    - Accessible by their creator during same connection
  - Global temporary (CREATE TABLE ##t)
    - Accessible by any user at any connection
  - Table variables (DECLARE @t TABLE)
    - Visible only to the connection that creates it
    - Deleted when out of scope

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- Tables are created using
  CREATE TABLE (preferred way)
  or
  SELECT . . . INTO statement
- DROP TABLE is used to delete a table
- ALTER TABLE is used to add, modify or delete columns

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Container for grouping tables, views, SP etc.

```
CREATE SCHEMA Sales;
GO

CREATE TABLE Sales.Customers(
    CustomerId INT          NOT NULL,
    Name       NVARCHAR(50) NOT NULL
);
```

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Table and Column Names

- *Requirements for standard identifiers:*
  - First character must be a letter or an underscore
  - Subsequent characters can be letters, digits, @, $, # and _
  - It must not be a T-SQL reserved word
  - Embedded spaces or special characters are not allowed

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- Any identifier that does not follow the standard identifier naming rules must be delimited using either quotation marks (“ “) or square brackets ([ ])

- SET QUOTED_IDENTIFIER ON must be used if using quotation marks (default is ON)

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Choosing Data Types Guideline

- Use the data type that requires the least amount of disk space
- It is costly to change data type later
- Use a variable-length data type, such as NVARCHAR or VARCHAR unless data is going to be updated frequently
- Avoid using the DATETIME and SMALLDATETIME because they use more disk space and provide less precision than the new DATE, TIME, and DATETIME2 data types.

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- Use the VARCHAR(max), NVARCHAR (max), and VARBINARY(max) data types instead of the TEXT, NTEXT, and IMAGE data types
  - Only use (max) if a specified size cannot be used
- Use the FLOAT or REAL data types only if the precision provided by decimal is insufficient.

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NULL or NOT NULL Constraint

- If the value for the column is optional - the column must allow NULLs
- Otherwise do not allow NULL

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Identity

- All tables should have one column or a combination of columns that uniquely identifies rows in the table
- This is called the PRIMARY KEY
- A primary key, must be guaranteed to be unique and their values must not change frequently

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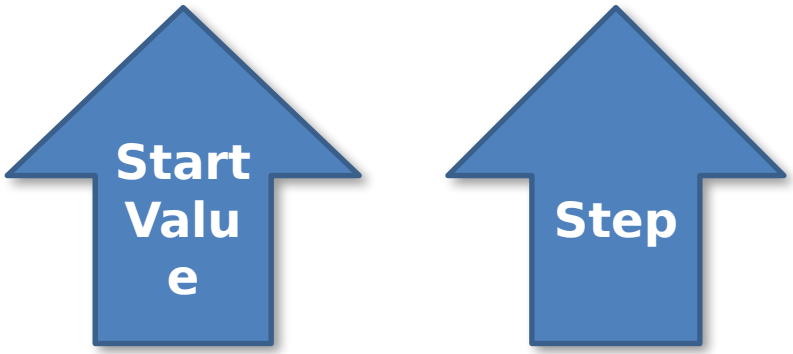
```
 <column name> <data type>
IDENTITY(<seed>, <increment>) NOT NULL
```



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```sql
CREATE TABLE HR.Employees (
    EmployeeId INT IDENTITY(1000,2)
                                NOT NULL
    ,FirstName NVARCHAR(50)  NOT NULL
    ,LastName  NVARCHAR(50)  NOT NULL
    ,BirthDate DATE          NOT NULL
);
```

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- IDENTITY columns can not allow NULL
- An existing column cannot be modified to use the IDENTITY property
- The IDENTITY column can be used only on data types that store whole numbers
- They are not guaranteed to generate complete sequences

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Sequence Numbers

- Are used to generate sequence numbers by using sequence objects
- Sequence objects are an optional way to generate unique numeric values in a table without many limitations of Identity
- A sequence is created independently of the tables by using the CREATE SEQUENCE

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Computed Columns

- A Computed Column is based on a formula that can reference the values of non computed columns in the same row of the same table
- Scalar functions can also be used in the definition of a computed column
- You can never write directly to

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```sql
CREATE TABLE OrderDetails
(
  orderid    INT            NOT NULL,
  unitprice  Decimal (8,2)  NOT NULL,
  qty        INT            NOT NULL,
  totalcost  AS unitprice * qty PERSISTED
);
```

Computed Column

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- PERSISTED keyword can be used to physically store the column values
- Only computed columns using deterministic functions can be marked as PERSISTED
- PERSISTED computed columns can be indexed

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Data Integrity

- Data integrity ensures that data entered into the database are:
  - Accurate,
  - Valid,
  - Consistent

- Any applicable integrity constraints and data validation rules must be satisfied before permitting a change to the database

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Validating Data

- Declarative data integrity: Implementing Constrains on tables and columns when defining them
- Procedural data integrity: Implementing Stored Procedures and/or Triggers to check data before and after DML statements

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Declarative Data Integrity

- *There are five type of constraints*

1. PRIMARY KEY
2. UNIQUE
3. FOREIGN KEY
4. CHECK
5. DEFAULT

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1) Primary key Constraints

- This constraints represent the unique identifier column/s that will enforce the uniqueness of each row

- It is a best practice to always have a PRIMARY KEY in each table and to use Identities that have no meaning to the application

- Primary Key cannot contain more than 16 columns or 900 bytes of

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- The default index that is created for a primary key is a clustered index

```
CREATE TABLE Production.Categories
(
  categoryid    INT            NOT NULL IDENTITY,
  categoryname NVARCHAR(15)   NOT NULL,
  description  NVARCHAR(200) NOT NULL,
  CONSTRAINT PK_Categories
      PRIMARY KEY (categoryid)
);
```

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```sql
-- Primary key as a clustered index.
ALTER TABLE MyTable
    ADD PRIMARY KEY (MyTableID);

-- Primary key as a nonclustered index.
ALTER TABLE MyTable
    ADD PRIMARY KEY NONCLUSTERED
(MyTableID);
```

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- sys.key_constraints and sys.indexes catalog views provide information on PKs

List the primary key constraints in a database

```
SELECT *
FROM   sys.key_constraints
WHERE  type = 'PK';
```

```
SELECT *
FROM   sys.indexes
WHERE  object_id =
OBJECT_ID('Production.Categories') AND
     name = 'PK_Categories';
```

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2) Unique constraints

- Unique constraints are similar to PRIMARY KEY constraints, except that you can have more than one unique constraint per table
- Unique constraint is a non-clustered index
- Does not require NOT NULL constraint

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3) FOREIGN KEY CONSTRAINTS

- Foreign key constraints identify a column or combination of columns whose values must exist in another column or combination of columns in another table in the same database

- These constraints enforce a relationship between two tables

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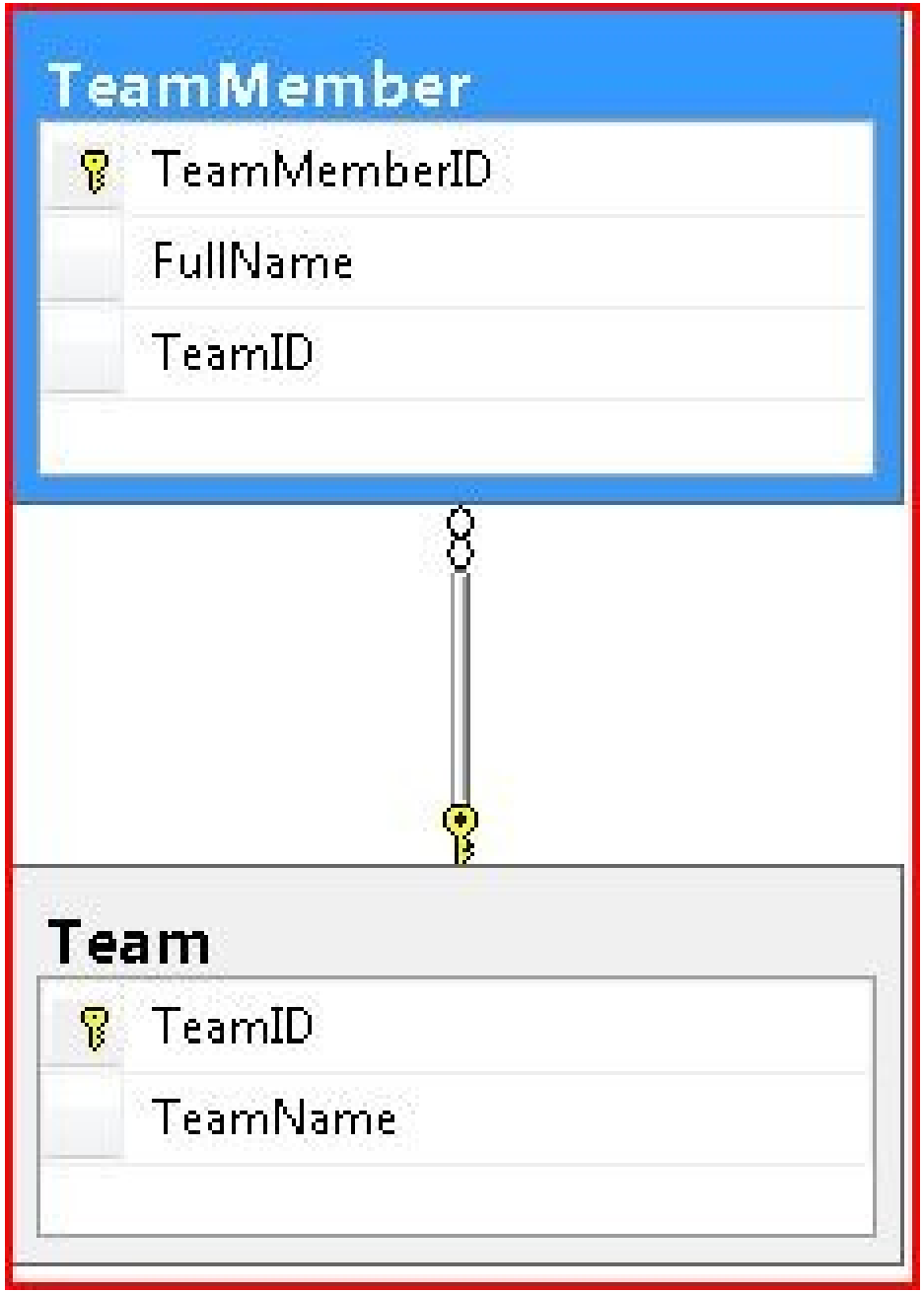

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```
CREATE TABLE Team(
   TeamID                int PRIMARY KEY,
   TeamName              varchar(50));


CREATE TABLE TeamMember(

   TeamMemberID int PRIMARY KEY,
   FullName     varchar(100),
   TeamID       int CONSTRAINT

FK_Team_TeamMem
   FOREIGN KEY REFERENCES Team(TeamID)
);
```

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- **Foreign key rules**:
  - The columns must have exactly the same data type and collation
  - The columns being referenced must have a unique index created on them. (primary key or unique constraint)
  - The foreign key columns have the same size limitations as that of the primary key and unique constraints

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- SQL Server queries FKs to verify referential integrity when deleting or updating primary key rows

- sys.foreign_keys and sys.foreign_key_columns catalog views provide information on FKs

```
SELECT *
FROM sys.foreign_keys
WHERE name = 'FK_Products_Categories';
```

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- A foreign key constraint stops a referential integrity violation which may occur because of a DELETE or an UPDATE of a row that it references, the default reaction is to raise an error message

- You can change the default action to:
  - NO ACTION (the default)
  - SET NULL
  - SET DEFAULT

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- No ACTION: This is the default action. No special action is taken, and if the FOREIGN KEY is violated, the statement rolls back
- CASCADE: Propagate the update or delete action to child rows
- SET NULL: Set the values of the FOREIGN KEY column to null for all related records
- SET DEFAULT: Set the values of the FOREIGN KEY column to its default values for all related records

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```
CREATE TABLE Test.Customers
(
    CustomerID INT PRIMARY KEY
);

CREATE TABLE Test.Orders
(
    OrderID INT PRIMARY KEY
    ,CustomerID INT NULL
 FOREIGN KEY REFERENCES Test.Customers
                ON DELETE SET NULL
                ON UPDATE CASCADE
);
```

Optional

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4) CHECK CONSTRAINTS

- These constraints validate the integrity of data in a column by checking it against a valid comparison
- They reject values that evaluate to False
- Error messages from check constraints are system-generated and cannot be replaced by a more

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- A check constraint cannot “see” the previous value of a column
- The expression in a check constraint can contain most of the logic that you can use in a WHERE clause
- It can call scalar UDFs and reference other columns in the same table
- It is not allowed to contain

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```sql
CREATE TABLE Stars (
   StarID int PRIMARY KEY,
   StarName varchar (50) Unique,
   SolarMass decimal (10,2)
      CHECK (SolarMass > 0),
   StarType varchar (50)
      DEFAULT 'Orange Giant'
);
GO

INSERT Stars(StarID, StarName, SolarMass,
StarType)
VALUES (3, 'Deneb', -6, 'supergiant');
```

Conflicts with check constraint

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Check & FK Constraints vs QUERY PERFORMANCE

- Because foreign keys and check constraints are declared rules, the optimizer can use them to create more efficient query plans

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